

# SRI SATYA VENKATA VINAY MAHESH PATNALA

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## PROFESSIONAL SUMMARY

Robotics and Electronics Engineer with an MSc from Heriot-Watt University and hands-on experience across the full robotics stack from embedded hardware and ROS to computer vision and machine learning. Built and deployed projects in autonomous manipulation, human-robot interaction, and AI-powered defect detection using industry-relevant tools including Python, PyTorch, COLMAP, and Arduino. Member of the Edinburgh Centre for Robotics and Heriot-Watt Maker Society. Looking for opportunities in robotics, automation and computer vision in the UK.

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## EDUCATION

**MSc in Robotics** | Heriot-Watt University, United Kingdom | 2025 - Present

- **Core Modules:** Robotic Mechanical Systems, Machine Vision, Robotic System Science, Intelligent Robotics, Human-Robot Interaction.
- **Dissertation: *Defect Detection in Complex Environments using NeRF and Gaussian Splatting*.** Defect Detection in Complex Environments using NeRF and 3D Gaussian Splatting supervised by Dr. Maria Koskinopoulou & Dr. Abderrahim Halimi. Reconstructed both surfaces of a physical Hard Disk Drive using Nerf studio, Splatfacto.
- **Active member of the Edinburgh Centre for Robotics (ECR) research community**

**B. Tech in Electronics and Communication Engineering** | Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Chennai, India | 2018 – 2022

- **Publication:** Presented "Design of Printed Antenna for Next Generation Wi-Fi Devices" at an International Conference (Chennai, May 2022).
  - **Focus:** Enthusiastic participant in workshops focusing on Arduino, Embedded programming, Artificial Intelligence, process development, and Prototyping.
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## TECHNICAL SKILLS

- **3D Reconstruction & Vision:** Neural Radiance Fields (NeRF), 3D Gaussian Splatting, COLMAP, Nerf studio, 3D Point Cloud Processing, Defect Detection, Machine Vision, Chamfer Distance.
  - **Robotics & Simulation:** ROS (Robot Operating System), Webots (Intelligent Robotics), Robot Kinematics, HRI Tools.
  - **Deep Learning & AI:** PyTorch, CNN Architectures, CUDA GPU Programming, Google Colab (GPU Cloud), Jupyter Notebooks
  - **Programming & Tools:** Python, C, MATLAB, Bash / Linux Shell Scripting, ffmpeg, Visual Studio
  - **Embedded & Hardware:** Arduino, ESP32, Chip Design Development, Embedded Systems.
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## KEY ACADEMIC PROJECTS

**Defect Detection via NeRF & Gaussian Splatting | MSc Dissertation**

*Heriot-Watt University Supervised by Dr. Maria Koskinopoulou & Dr. Abderrahim Halimi*

- Built an end-to-end pipeline: raw video → ffmpeg extraction → COLMAP pose estimation → Nerfstudio training (Nerfacto, Splatfacto) → point cloud export → defect analysis.
- Implementing and benchmarking three neural rendering approaches **Nerfacto** (volumetric NeRF), **3D Gaussian Splatting (Splatfacto)** evaluated on PSNR, SSIM, LPIPS and real-time FPS metrics.
- Preprocessing raw video frames using ffmpeg for frame extraction and filtering, applying image quality checks to ensure optimal input for COLMAP pose estimation and consistent model training.
- Cross-validated 3D results against a 2D morphological top-hat MATLAB pipeline, confirming 135° dominant scratch angle and 169 mm maximum scratch length.

#### **4-DOF Autonomous Robotic Arm** | *Robotic Mechanical Systems*

**Heriot-Watt University Supervised by Dr. Xianwen Kong**

- Designed and 3D-printed a 4-DOF robotic arm; implemented full forward/inverse kinematics pipeline in MATLAB.
- Integrated OpenCV-based vision system for object detection and localisation, enabling closed-loop pick-and-place operations with Arduino actuation.

#### **COSMIC: AI Astronaut Companion** | *Conversational Agents*

**Heriot-Watt University Supervised by Dr. Alessandro Suglia, Dr. Marta Romeo**

- Engineered a voice-activated AI using **React**, **Python**, and **OpenAI Whisper**.
- Integrated Wav2Vec2 emotion detection to generate empathetic, context-aware responses via LLM demonstrating multimodal HRI design.

#### **Cheer Mate: HRI System** | *Human-Robot Interaction*

**Heriot-Watt University Supervised by Dr. Christian Dondrup**

- Programmed a Marty V2 robot as an interactive fitness coach using Python and Ollama LLM for natural language dialogue.
- Developed multimodal feedback system (speech + physical movement) and QR-based activity tracking for personalised coaching sessions.

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## **PROFESSIONAL EXPERIENCE**

**Python & Data Science Intern** | EI Systems Services, India | *Feb 2024 – April 2024*

- Completed comprehensive training on Python with a focus on Data Science, acquiring skills in data manipulation, analysis, and visualization.
- Utilized machine learning algorithms to predict insurance claims, enhancing the decision-making process.
- Developed skills in coding, data analysis, cybersecurity, and forensic technology principles

**Kitchen Assistant** | Pasta Evangelists, United Kingdom | *April 2025 – Present*

- Maintain high operational standards in a high-volume, fast-paced kitchen environment while balancing full-time MSc studies.
- Collaborate effectively within a diverse team to ensure efficient workflow and adherence to strict health and safety regulations.
- Demonstrate reliability, time management, and the ability to work under pressure to meet production targets.

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## **PUBLICATIONS & ACHIEVEMENTS**

- Conference Paper: "Design of Printed Antenna for Next Generation Wi-Fi Devices" International Conference on Emerging Technologies, Chennai, May 2022.
- Edinburgh Centre for Robotics (ECR) affiliated researcher through Heriot-Watt University, collaborating within one of the UK's leading robotics research hubs.
- Actively contributing dissertation findings to potential publication on NeRF-based industrial inspection pipelines.

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## **EXTRACURRICULAR ACTIVITIES & LEADERSHIP**

- Active participant in weekly Makerspace sessions, collaborating with cross-disciplinary students to prototype and build hardware/software projects.
- Utilise society equipment including 3D printers, laser cutters, and resin printers to rapidly prototype components for robotics coursework and personal projects, complementing dissertation research.
- Engage in skill-building workshops on 3D modelling and Arduino, and contribute to inter-society collaborative projects, strengthening both technical and teamwork competencies beyond the classroom.